

Elements

An element is a substance made up of just one type of atom. Scientists have discovered 117 different elements. The chart on this page, called the periodic table, shows most of them.

The elements in our bodies mostly come from what we eat.



Milk contains the element calcium, which helps form your teeth and bones.



This bucket is made of the element iron, coated with zinc, which stops iron from rusting.

The periodic table

In the periodic table, elements are arranged by the number of protons in their atoms, starting with hydrogen. Elements with similar properties fall into groups, which are shown in color.

1 H HYDROGEN 1	2 Li LITHIUM 3	Be BERYLLIUM 4							
Na SODIUM 11	Mg MAGNESIUM 12		3	4	5	6	7	8	9
K POTASSIUM 19	Ca CALCIUM 20	Sc SCANDIUM 21	Ti TITANIUM 22	V VANADIUM 23	Cr CHROMIUM 24	Mn MANGANESE 25	Fe IRON 26	Co COBALT 27	
Rb RUBIDIUM 37	Sr STRONTIUM 38	Y YTTRIUM 39	Zr ZIRCONIUM 40	Nb NIOBIUM 41	Mo MOLYBDENUM 42	Tc TECHNETIUM 43	Ru RUTHENIUM 44	Rh RHODIUM 45	
Cs CAESIUM 55	Ba BARIUM 56	LANTHANIDES or RARE-EARTH METALS 57—71	Hf HAFNIUM 72	Ta TANTALUM 73	W TUNGSTEN 74	Re RHENIUM 75	Os OSMIUM 76	Ir IRIDIUM 77	
Fr FRANCIUM 87	Ra RADIUM 88	ACTINIDES or RARE-EARTH RADIOACTIVE METALS 89—103	Rf RUTHERFORDIUM 104	Db DUBNIUM 105	Sg SEABORGIUM 106	Bh BOHRIUM 107	Hs HASSIUM 108	Mt MEITNERIUM 109	
			La LANTHANUM 57	Ce CERIUM 58	Pr PRASEODYMIUM 59	Nd NEODYMIUM 60	Pm PROMETHIUM 61	Sm SAMARIUM 62	
			Ac ACTINIUM 89	Th THORIUM 90	Pa PROTACTINIUM 91	U URANIUM 92	Np NEPTUNIUM 93	Pu PLUTONIUM 94	

Each vertical column is called a GROUP, or family, of elements. Some groups have elements sharing very similar properties. Other groups have elements with less in common.

Metal and non-metals

Most elements are metals, and the others are called non-metals. Metals are normally solid, shiny, and hard. They all conduct electricity and heat. Silver, aluminum, and zinc are metals. Carbon, oxygen, and silicon are non-metals.